PHASE 2 — TECHNICAL APPLICATION

Baseline Implementation: Diabetes Prediction Using Machine Learning and Explainable AI Techniques

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1. Introduction and objective

This phase implements a practical baseline for the research paper by Tasin, Nabil, Islam and Khan (2023). The purpose is to connect the methodological issues identified in Phase 1 with a reproducible machine-learning experiment. Two conventional classifiers—Logistic Regression and Random Forest—are trained and evaluated using Python and scikit-learn on the publicly available Pima Indians Diabetes Database.

The research question is: how well do simple Logistic Regression and Random Forest models classify diabetes outcomes on the Pima dataset when data preprocessing is fitted only on the training partition? The implementation is a baseline evaluation, not a clinical diagnostic system.

2. Dataset and target definition

The Pima Indians Diabetes Database contains 768 records, eight numerical predictors and a binary target named
Outcome. The predictors are Pregnancies, Glucose, BloodPressure, SkinThickness, Insulin, BMI, DiabetesPedigreeFunction and Age. Outcome = 1 indicates diabetes and Outcome = 0 indicates no diabetes. The dataset is publicly available as a CSV file through a commonly used public mirror.

The paper also uses a separate RTML dataset containing 203 female garment workers from Bangladesh. The Phase 2 baseline deliberately uses the Pima dataset alone because it is publicly accessible and permits reproducible implementation. Consequently, its metrics are not directly comparable to the paper's combined-dataset results.

3. Data preparation and preprocessing

The implementation first checks the target distribution and counts zero values in the physiological measurements. As discussed in Phase 1, zero is a legitimate value for Pregnancies but is generally not a plausible measurement for Glucose, BloodPressure, SkinThickness, Insulin or BMI. These zero values are therefore converted to missing values before modelling. The notebook records their counts and percentages.

A stratified 80:20 train/test split is created with random_state = 42. Stratification preserves approximately the same outcome proportions in both partitions. Median imputation is fitted inside each model's scikit-learn pipeline, so the median values are learned from training data and then applied to the test data. Logistic Regression additionally uses StandardScaler, also within its pipeline. Random Forest does not require standardisation because it is tree-based. No oversampling is applied in this baseline; this enables an initial comparison without synthetic samples.

4. Baseline algorithms

Logistic Regression. Logistic Regression is a supervised binary classification algorithm that estimates the probability of a positive class through a logistic function. It is included as an interpretable, computationally efficient baseline. Since its optimisation is sensitive to feature scale, the implementation standardises the imputed predictors.

Random Forest. Random Forest combines predictions from multiple decision trees trained using randomised samples and feature choices. It can model nonlinear relationships and interactions between predictors and generally does not require feature scaling. The notebook uses 300 trees, a fixed random seed and a minimum leaf size of two.

5. Evaluation strategy and metrics

Both models are trained on the training partition and evaluated on the same untouched test partition. The notebook reports accuracy, precision, recall (sensitivity), F1-score and ROC-AUC, alongside classification reports, confusion matrices and ROC curves. Accuracy summarises the proportion of correct predictions, while precision measures the proportion of predicted positive cases that are positive. Recall measures the proportion of actual positive cases identified, and F1 is the harmonic mean of precision and recall. ROC-AUC evaluates ranking ability across classification thresholds.

Using several metrics is important because the dataset is class-imbalanced and the consequences of false negatives may differ from those of false positives. These metrics alone do not establish clinical usefulness; threshold selection, calibration, external validation and clinical review would be needed for any real screening application.

6. Comparison with the research paper

Approach	Accuracy	F1-score	ROC-AUC
Paper: XGBoost + ADASYN (headline result)	0.81	0.81	0.84
Paper: XGBoost + SMOTE (results table)	0.78	0.78	0.84
Logistic Regression baseline	0.7143	0.5686	0.8178
Random Forest baseline	0.7532	0.6122	0.8113

The paper reports a headline result of 81% accuracy, F1 = 0.81 and AUC = 0.84 for XGBoost with ADASYN. It also reports 78% accuracy, F1 = 0.78 and AUC = 0.84 for XGBoost with SMOTE. Phase 1 noted an inconsistency in the article: a comparison table lists 88.5% accuracy for the ADASYN model, while the abstract/results report 81%. This report uses the headline result and explicitly preserves the inconsistency rather than treating both values as equivalent.

7. Expected methodological lessons and limitations

- A baseline provides a reference point for assessing whether a more complex approach is justified.
- Converting impossible zeros to missing values is more defensible than treating them as genuine physiologicalmeasurements.
- Fitting imputation and scaling within a pipeline reduces the risk that test-set information influences preprocessing.

- A fixed test split makes the experiment reproducible, but its result can vary with the split. Repeated or nestedcross-validation would provide a stronger estimate of generalisation.
- The experiment is restricted to the Pima dataset and should not be generalised to other populations withoutexternal validation.
- This dataset and model are for coursework only. The results must not be used to diagnose an individual.

8. Conclusion

This practical phase establishes a reproducible baseline workflow for diabetes classification using Logistic Regression and Random Forest. It addresses key concerns from Phase 1 by handling physiologically invalid zeros as missing values and fitting imputation and scaling only within the training pipeline. The actual measured metrics are generated by the accompanying notebook and must be inserted into the comparison table after execution. The paper's reported XGBoost + ADASYN result provides context, but a controlled comparison would require matching the dataset, split strategy, preprocessing, and evaluation procedure.

9. GitHub Repository URL: https://github.com/MMaleesha/Diabetes-Prediction-Using-Machine-Learning-and-Explainable-AI-Techniques

10. References

Tasin, I., Nabil, T. U., Islam, S., & Khan, R. (2023). Diabetes prediction using machine learning and explainable AI techniques. *Healthcare Technology Letters, 10*(1–2), 1–10. https://doi.org/10.1049/htl2.12039

Smith, J. W., Everhart, J. E., Dickson, W. C., Knowler, W. C., & Johannes, R. S. (1988). Using the ADAP learning algorithm to forecast the onset of diabetes mellitus. *Proceedings of the Annual Symposium on Computer Applications in Medical Care*, 261–265.

Dataset mirror: https://github.com/jbrownlee/Datasets/blob/master/pima-indians-diabetes.data.csv